Solarpanel Location Data Analyser

Data Analytics Assignment

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1 Running Code

Library Installing

Also disables all warning messages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(cowplot)
```

```
Warning: package 'cowplot' was built under R version 4.5.3


Attaching package: 'cowplot'

The following object is masked from 'package:lubridate':

    stamp
```

```
library(ggplot2)
library(ggrepel)
```

```
Warning: package 'ggrepel' was built under R version 4.5.3
```

```
options(warn = -1)
```

Dataset Installing

This code block downloads all the required CSV files from the project directory. For this to work ensure all CSV files are in same folder location as the R file.

This code also;

- Bounds the data to only include data that is >= 1995.
- Create vectors of yearly rainfall and no.days of rain per year
- Ensures all rain, solar and temperature vectors are off the same length so they can be concatenated in later sections. This is done by adding null (NA) cells to the vectors to ensure they all the same lengths. For the rain variables, this is done either first or last depending of what years are missing.

```
## Ernabella (replacement for Fregon) ##

Ernabella_Solar <- read.csv("Ernabella Solar Exposure - IDCJAC0016_016097_1800_Data.csv")
Ernabella_Rain <- read.csv("Ernabella Rainfall - IDCJAC0009_016097_1800_Data.csv")
Ernabella_Temp <- read.csv("Ernabella Max Temp - IDCJAC0010_016097_1800_Data.csv")

Ernabella_Rain <- filter(Ernabella_Rain,Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))

Ernabella_Solar <- filter(Ernabella_Solar,Year>1994)
Ernabella_Temp <- filter(Ernabella_Temp,Year>1994)

Ernabella_Solar_data <- c(Ernabella_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430

Ernabella_Temp_data <- c(Ernabella_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length

null_rain <- data.frame(Year=NA,Rainfall=NA,Days_Rain=NA)
```

```r
Ernabella_Rain <- rbind(null_rain,null_rain,Ernabella_Rain)

## Marla (Dallirina replacement)  ##

Marla_Solar <- read.csv("Marla Solar Exposure - IDCJAC0016_016085_1800_Data.csv")
Marla_Rain <- read.csv("Marla Rainfall - IDCJAC0009_016085_1800_Data.csv")
Marla_Temp <- read.csv("Marla Max Temp - IDCJAC0010_016085_1800_Data.csv")

Marla_Rain <- filter(Marla_Rain,Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Marla_Solar <- filter(Marla_Solar,Year>1994)
Marla_Temp <- filter(Marla_Temp,Year>1994)

Marla_Solar_data <- c(Marla_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-length

Marla_Temp_data <- c(Marla_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Marla_Te

Marla_Rain <- rbind(Marla_Rain, null_rain, null_rain,null_rain,null_rain,null_rain,null_rain

## Moomba (Mungerine Replacement)

Moomba_Solar <- read.csv("Moomba Solar Exposure - IDCJAC0016_017123_1800_Data.csv")
Moomba_Rain <- read.csv("Moomba Rainfall - IDCJAC0009_017123_1800_Data.csv")
Moomba_Temp <- read.csv("Moomba Max Temp - IDCJAC0010_017123_1800_Data.csv")

Moomba_Rain <- filter(Moomba_Rain,Year>1994)%>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Moomba_Solar <- filter(Moomba_Solar,Year>1994)
Moomba_Temp <- filter(Moomba_Temp,Year>1994)

Moomba_Solar_data <- c(Moomba_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-leng

Moomba_Temp_data <- c(Moomba_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Moomba

## Oodanata (Nilpinna Replacment)
```

```r
Oodanata_Solar <- read.csv("Oodanata Solar Exposure - IDCJAC0016_017043_1800_Data.csv")
Oodanata_Rain <- read.csv("Oodanata Rainfall - IDCJAC0009_017043_1800_Data.csv")
Oodanata_Temp <- read.csv("Oodanata Max Temp - IDCJAC0010_017043_1800_Data.csv")

Oodanata_Rain <- filter(Oodanata_Rain,Year>1994)%>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Oodanata_Solar <- filter(Oodanata_Solar,Year>1994)
Oodanata_Temp <- filter(Oodanata_Temp,Year>1994)

Oodanata_Solar_data <- c(Oodanata_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-l

Oodanata_Temp_data <- c(Oodanata_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Oo

## Arkaroola (Wymbanna Replacement)

Arkaroola_Solar <- read.csv("Arkaroola Solar Exposure - IDCJAC0016_017099_1800_Data.csv")
Arkaroola_Rain <- read.csv("Arkaroola Rainfall - IDCJAC0009_017099_1800_Data.csv")
Arkaroola_Temp <- read.csv("Arkaroola Max Temp - IDCJAC0010_017099_1800_Data.csv")

Arkaroola_Rain <- filter(Arkaroola_Rain,Year>1994)%>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Arkaroola_Solar <- filter(Arkaroola_Solar,Year>1994)
Arkaroola_Temp <- filter(Arkaroola_Temp,Year>1994)

Arkaroola_Solar_data <- c(Arkaroola_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 1143

Arkaroola_Temp_data <- c(Arkaroola_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length

## Tarcoola (Wirraminna Replacement)

Tarcoola_Solar <- read.csv("Tarcoola Solar Exposure - IDCJAC0016_016098_1800_Data.csv")
Tarcoola_Rain <- read.csv("Tarcoola Rainfall - IDCJAC0009_016098_1800_Data.csv")
Tarcoola_Temp <- read.csv("Tarcoola Max Temp - IDCJAC0010_016098_1800_Data.csv")

Tarcoola_Rain <- filter(Tarcoola_Rain,Year>1994)%>%
  group_by(Year) %>%
```

```
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Tarcoola_Solar <- filter(Tarcoola_Solar,Year>1994)
Tarcoola_Temp <- filter(Tarcoola_Temp,Year>1994)

Tarcoola_Solar_data <- c(Tarcoola_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-l

Tarcoola_Temp_data <- c(Tarcoola_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Ta

Tarcoola_Rain <- rbind(null_rain,null_rain,Tarcoola_Rain)

## Pirie (Hallet Replacement)

Pirie_Solar <- read.csv("Pirie Solar Exposure - IDCJAC0016_021043_1800_Data.csv")
Pirie_Rain <- read.csv("Pirie Rainfall - IDCJAC0009_021043_1800_Data.csv")
Pirie_Temp <- read.csv("Pirie Max Temp - IDCJAC0010_021043_1800_Data.csv")

Pirie_Rain <- filter(Pirie_Rain,Year>1994)%>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Pirie_Solar <- filter(Pirie_Solar,Year>1994)
Pirie_Temp <- filter(Pirie_Temp,Year>1994)

Pirie_Solar_data <- c(Pirie_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-length

Pirie_Temp_data <- c(Pirie_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Pirie_Te

Pirie_Rain <- rbind(Pirie_Rain, null_rain,null_rain,null_rain,null_rain,null_rain,null_rain,n
## Naracoorte (Coonawarra Replacement)

Naracoorte_Solar <- read.csv("Naracoorte Solar Exposure - IDCJAC0016_026099_1800_Data.csv")
Naracoorte_Rain <- read.csv("Naracoorte Rainfall - IDCJAC0009_026099_1800_Data.csv")
Naracoorte_Temp <- read.csv("Naracoorte Max Temp - IDCJAC0010_026099_1800_Data.csv")

Naracoorte_Rain <- filter(Naracoorte_Rain,Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
```

```
Naracoorte_Solar <- filter(Naracoorte_Solar,Year>1994)
Naracoorte_Temp <- filter(Naracoorte_Temp,Year>1994)

Naracoorte_Solar_data <- c(Naracoorte_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 114

Naracoorte_Temp_data <- c(Naracoorte_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-lengt

Naracoorte_Rain <- rbind(Naracoorte_Rain,null_rain,null_rain,null_rain)

# Ceduna

Ceduna_Solar <- read.csv("Ceduna Solar Exposure - IDCJAC0016_018012_1800_Data.csv")
Ceduna_Rain <- read.csv("Ceduna Rainfall - IDCJAC0009_018012_1800_Data.csv")
Ceduna_Temp <- read.csv("Ceduna Max Temp - IDCJAC0010_018012_1800_Data.csv")

Ceduna_Solar <- filter(Ceduna_Solar,Year>1994)
Ceduna_Rain <- filter(Ceduna_Rain,Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Ceduna_Temp <- filter(Ceduna_Temp,Year>1994)

Ceduna_Solar_data <- c(Ceduna_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-lengt

Ceduna_Temp_data <- c(Ceduna_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Ceduna

# Roxby
Roxby_solar <- read.csv("Roxby Solar Exposure - IDCJAC0016_016096_1800_Data.csv")
Roxby_rain <- read.csv("Roxby Rainfall - IDCJAC0009_016096_1800_Data.csv")
Roxby_Temp <- read.csv("Roxby Max Temp - IDCJAC0010_016096_1800_Data.csv")

Roxby_solar <- filter(Roxby_solar, Year>1994)
Roxby_rain <- filter(Roxby_rain, Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Roxby_Temp <- filter(Roxby_Temp,Year>1994)

Roxby_Solar_data <- c(Roxby_solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-length
```

```
Roxby_temp_data <- c(Roxby_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Roxby_Te

Roxby_rain <- rbind(Roxby_rain,null_rain,null_rain)

# Padthaway

Padthaway_solar <- read.csv("Padthaway Solar Exposure - IDCJAC0016_026100_1800_Data.csv")
Padthaway_rain <- read.csv("Padthaway Rainfall - IDCJAC0009_026100_1800_Data.csv")
Padthaway_Temp <- read.csv("Padthaway Max Temp - IDCJAC0010_026100_1800_Data.csv")

Padthaway_solar <- filter(Padthaway_solar, Year>1994)
Padthaway_rain <- filter(Padthaway_rain, Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Padthaway_Temp <- filter(Padthaway_Temp,Year>1994)

Padthaway_Solar_data <- c(Padthaway_solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430

Padthaway_Temp_data <- c(Padthaway_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length

Padthaway_rain <- rbind(Padthaway_rain,null_rain,null_rain,null_rain,null_rain,null_rain)

# Cleve

Cleve_Solar <- read.csv("Cleve Solar Exposure - IDCJAC0016_018014_1800_Data.csv")
Cleve_Rain <- read.csv("Cleve Rainfall - IDCJAC0009_018014_1800_Data.csv")
Cleve_Temp <- read.csv("Cleve Max Temp - IDCJAC0010_018014_1800_Data.csv")

Cleve_Solar <- filter(Cleve_Solar, Year>1994)
Cleve_Rain <- filter(Cleve_Rain, Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))
Cleve_Temp <- filter(Cleve_Temp,Year>1994)

Cleve_Solar_data <- c(Cleve_Solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-length

Cleve_Temp_data <- c(Cleve_Temp$Maximum.temperature..Degree.C., rep(NA, 11430-length(Cleve_Te
```

```
# Mulka

Mulka_solar <- read.csv("Mulka Solar Exposure - IDCJAC0016_017036_1800_Data.csv")
Mulka_rain <- read.csv("Mulka Rainfall - IDCJAC0009_017036_1800_Data.csv")
# Mulka has no temp csv

Mulka_solar <- filter(Mulka_solar, Year>1994)
Mulka_rain <- filter(Mulka_rain, Year>1994) %>%
  group_by(Year) %>%
  summarise(
      Rainfall = sum(Rainfall.amount..millimetres., na.rm = TRUE),
      Days_Rain = sum(Rainfall.amount..millimetres. > 0, na.rm = TRUE))


Mulka_Solar_data <- c(Mulka_solar$Daily.global.solar.exposure..MJ.m.m., rep(NA, 11430-length

Mulka_temp_data <- c(rep(NA, 11430))

Mulka_rain <- rbind(Mulka_rain, null_rain,null_rain,null_rain,null_rain,null_rain,null_rain,n
```

Data Concatenation

This code block joins the respective 4 variables together from each station. This also creates a vector containing all the median values for each variable for each station.

```
Solar_Joined <- data.frame(Ernabella = Ernabella_Solar_data, Marla = Marla_Solar_data, Moomba

Rain_Joined <- data.frame(Ernabella = Ernabella_Rain$Rainfall, Marla = Marla_Rain$Rainfall, M

Raining_Joined <- data.frame(Ernabella = Ernabella_Rain$Days_Rain, Marla = Marla_Rain$Days_Ra

Temp_Joined <- data.frame(Ernabella = Ernabella_Temp_data, Marla = Marla_Temp_data, Moomba =

Solar_med <- apply(Solar_Joined, 2, median, na.rm = TRUE)
Rain_med <- apply(Rain_Joined, 2, median, na.rm=TRUE)
Temp_med <- apply(Temp_Joined, 2, median, na.rm = TRUE)
Raining_med <- apply(Raining_Joined, 2, median, na.rm=TRUE)

median_values <- data.frame(Solar=Solar_med,Rain=Rain_med,Temp = Temp_med, Raining = Raining_

median_values <- rownames_to_column(median_values, var = "Location")
```

Hypothesis Testing

Plots QQ diagrams for all stations using solar variable.

Runs Krusal Wallis test

```
Solar_Joined_Long <- Solar_Joined %>%
  pivot_longer( cols = everything(), names_to = "Stations", values_to = "Solar")

Anova_Solar <- aov(Solar~Stations, data = Solar_Joined_Long)
summary(Anova_Solar)
```

```
                Df  Sum Sq Mean Sq F value Pr(>F)    
Stations        12  438158   36513   616.4 <2e-16 ***
Residuals   145464 8617268      59                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
3113 observations deleted due to missingness
```

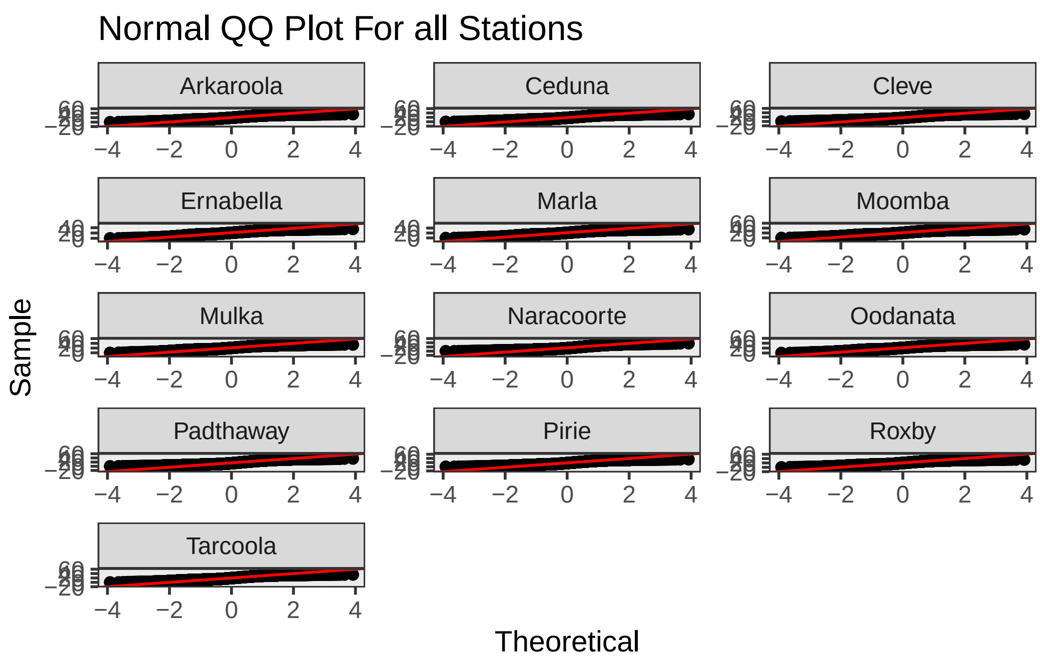
```
TukeyHSD(Anova_Solar)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = Solar ~ Stations, data = Solar_Joined_Long)

$Stations
                             diff         lwr         upr     p adj
Ceduna-Arkaroola     -1.341604921 -1.68250220 -1.00070764 0.0000000
Cleve-Arkaroola      -2.542130584 -2.88302023 -2.20124093 0.0000000
Ernabella-Arkaroola   0.869386662  0.52867986  1.21009346 0.0000000
Marla-Arkaroola       0.818944737  0.47823794  1.15965154 0.0000000
Moomba-Arkaroola      0.886189012  0.54547461  1.22690342 0.0000000
Mulka-Arkaroola       0.894316248  0.55344949  1.23518300 0.0000000
Naracoorte-Arkaroola -4.025937595 -4.36666722 -3.68520798 0.0000000
Oodanata-Arkaroola    0.988510441  0.64779603  1.32922485 0.0000000
Padthaway-Arkaroola  -3.722573067 -4.06347035 -3.38167578 0.0000000
Pirie-Arkaroola      -1.728300233 -2.06906791 -1.38753256 0.0000000
Roxby-Arkaroola       0.140775588 -0.20010643  0.48165761 0.9788975
Tarcoola-Arkaroola   -0.229270750 -0.57001559  0.11147409 0.5696758
Cleve-Ceduna         -1.200525663 -1.54160569 -0.85944563 0.0000000
Ernabella-Ceduna      2.210991583  1.87009430  2.55188887 0.0000000
```

Marla-Ceduna	2.160549658	1.81965238	2.50144694	0.0000000
Moomba-Ceduna	2.227793933	1.88688905	2.56869882	0.0000000
Mulka-Ceduna	2.235921170	1.89486402	2.57697832	0.0000000
Naracoorte-Ceduna	-2.684332674	-3.02525276	-2.34341258	0.0000000
Oodanata-Ceduna	2.330115362	1.98921048	2.67102025	0.0000000
Padthaway-Ceduna	-2.380968146	-2.72205580	-2.03988049	0.0000000
Pirie-Ceduna	-0.386695312	-0.72765343	-0.04573719	0.0109446
Roxby-Ceduna	1.482380510	1.14130811	1.82345291	0.0000000
Tarcoola-Ceduna	1.112334171	0.77139887	1.45326947	0.0000000
Ernabella-Cleve	3.411517246	3.07062760	3.75240690	0.0000000
Marla-Cleve	3.361075321	3.02018567	3.70196497	0.0000000
Moomba-Cleve	3.428319596	3.08742235	3.76921685	0.0000000
Mulka-Cleve	3.436446832	3.09539731	3.77749635	0.0000000
Naracoorte-Cleve	-1.483807011	-1.82471947	-1.14289456	0.0000000
Oodanata-Cleve	3.530641025	3.18974377	3.87153828	0.0000000
Padthaway-Cleve	-1.180442483	-1.52152251	-0.83936245	0.0000000
Pirie-Cleve	0.813830351	0.47287986	1.15478084	0.0000000
Roxby-Cleve	2.682906172	2.34184140	3.02397094	0.0000000
Tarcoola-Cleve	2.312859834	1.97193217	2.65378750	0.0000000
Marla-Ernabella	-0.050441925	-0.39114873	0.29026488	0.9999994
Moomba-Ernabella	0.016802350	-0.32391206	0.35751676	1.0000000
Mulka-Ernabella	0.024929586	-0.31593717	0.36579634	1.0000000
Naracoorte-Ernabella	-4.895324257	-5.23605388	-4.55459464	0.0000000
Oodanata-Ernabella	0.119123779	-0.22159063	0.45983819	0.9950374
Padthaway-Ernabella	-4.591959729	-4.93285701	-4.25106245	0.0000000
Pirie-Ernabella	-2.597686895	-2.93845457	-2.25691922	0.0000000
Roxby-Ernabella	-0.728611073	-1.06949309	-0.38772906	0.0000000
Tarcoola-Ernabella	-1.098657412	-1.43940225	-0.75791257	0.0000000
Moomba-Marla	0.067244275	-0.27347013	0.40795868	0.9999859
Mulka-Marla	0.075371511	-0.26549524	0.41623827	0.9999513
Naracoorte-Marla	-4.844882332	-5.18561195	-4.50415271	0.0000000
Oodanata-Marla	0.169565704	-0.17114870	0.51028011	0.9144695
Padthaway-Marla	-4.541517804	-4.88241509	-4.20062052	0.0000000
Pirie-Marla	-2.547244970	-2.88801265	-2.20647730	0.0000000
Roxby-Marla	-0.678169149	-1.01905117	-0.33728713	0.0000000
Tarcoola-Marla	-1.048215487	-1.38896033	-0.70747065	0.0000000
Mulka-Moomba	0.008127236	-0.33274712	0.34900159	1.0000000
Naracoorte-Moomba	-4.912126607	-5.25286383	-4.57138938	0.0000000
Oodanata-Moomba	0.102321429	-0.23840058	0.44304344	0.9988341
Padthaway-Moomba	-4.608762079	-4.94966696	-4.26785720	0.0000000
Pirie-Moomba	-2.614489246	-2.95526452	-2.27371397	0.0000000
Roxby-Moomba	-0.745413424	-1.08630304	-0.40452381	0.0000000
Tarcoola-Moomba	-1.115459762	-1.45621221	-0.77470732	0.0000000

```
Naracoorte-Mulka     -4.920253844 -5.26114341 -4.57936428 0.0000000
Oodanata-Mulka        0.094194192 -0.24668017  0.43506855 0.9994934
Padthaway-Mulka      -4.616889316 -4.95794646 -4.27583217 0.0000000
Pirie-Mulka          -2.622616482 -2.96354408 -2.28168888 0.0000000
Roxby-Mulka          -0.753540660 -1.09458255 -0.41249877 0.0000000
Tarcoola-Mulka       -1.123586999 -1.46449177 -0.78268222 0.0000000
Oodanata-Naracoorte   5.014448036  4.67371081  5.35518526 0.0000000
Padthaway-Naracoorte  0.303364528 -0.03755556  0.64428462 0.1408723
Pirie-Naracoorte      2.297637362  1.95684687  2.63842785 0.0000000
Roxby-Naracoorte      4.166713184  3.82580836  4.50761801 0.0000000
Tarcoola-Naracoorte   3.796666845  3.45589919  4.13743450 0.0000000
Padthaway-Oodanata   -4.711083508 -5.05198839 -4.37017862 0.0000000
Pirie-Oodanata       -2.716810674 -3.05758595 -2.37603540 0.0000000
Roxby-Oodanata       -0.847734852 -1.18862447 -0.50684523 0.0000000
Tarcoola-Oodanata    -1.217781191 -1.55853363 -0.87702875 0.0000000
Pirie-Padthaway       1.994272834  1.65331471  2.33523096 0.0000000
Roxby-Padthaway       3.863348656  3.52227625  4.20442106 0.0000000
Tarcoola-Padthaway    3.493302317  3.15236702  3.83423762 0.0000000
Roxby-Pirie           1.869075822  1.52813296  2.21001868 0.0000000
Tarcoola-Pirie        1.499029483  1.15822378  1.83983519 0.0000000
Tarcoola-Roxby       -0.370046339 -0.71096637 -0.02912630 0.0195952
```

```r
ggplot(Anova_Solar, aes(sample = Solar, group = Stations)) +
  stat_qq(distribution = stats::qnorm) +
  stat_qq_line(distribution = stats::qnorm, color = "red") +
  theme_bw() +    labs(x = "Theoretical", y = "Sample") +
  facet_wrap(~ Stations, ncol = 3, scales='free') +
  ggtitle("Normal QQ Plot For all Stations") +
  labs(x = "Theoretical", y = "Sample")
```




```r
Anova_Solar_Sum <-Solar_Joined_Long  %>%
  group_by(Stations) %>%
  summarise(n = n(),
            mean = mean(Solar,na.rm=TRUE),
            sd = sd(Solar,na.rm=TRUE))

r <- max(Anova_Solar_Sum$sd)/min(Anova_Solar_Sum$sd)
kruskal.test(Solar~Stations,data = Solar_Joined_Long)
```

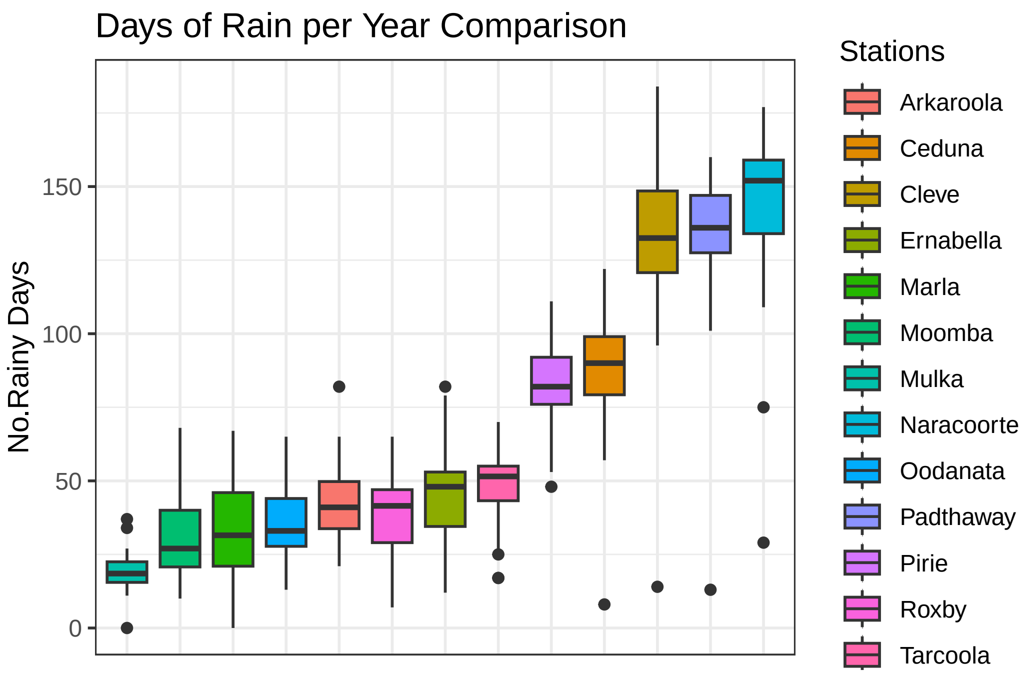
```
    Kruskal-Wallis rank sum test

data:  Solar by Stations
Kruskal-Wallis chi-squared = 7338.8, df = 12, p-value < 2.2e-16
```

Box Plots

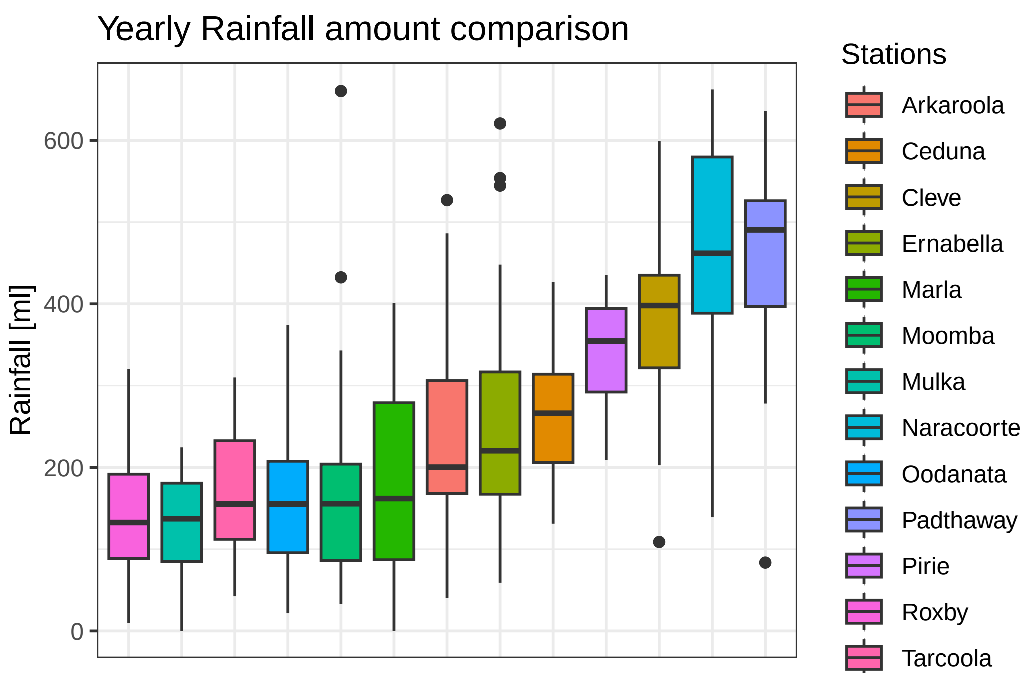
This code block displays the boxplots for all 4 variables.

```r
Solar_Plotting <- Solar_Joined %>%
  pivot_longer( cols = everything(), names_to = "Stations", values_to = "Solar Exposure Data'
```

```
Rain_Plotting <- Rain_Joined %>%
  pivot_longer( cols = everything(), names_to = "Stations", values_to = "Daily Rainfall amou

Raining_Plotting <- Raining_Joined %>%
  pivot_longer( cols = everything(), names_to = "Stations", values_to = "Days of Rain")

Temp_Plotting <- Temp_Joined %>%
  pivot_longer( cols = everything(), names_to = "Stations", values_to = "Max Daily Temperatu


ggplot(Raining_Plotting, aes(x = reorder(Stations, `Days of Rain`, FUN = median, na.rm = TRUE
  geom_boxplot() +
  labs(
    title = "Days of Rain per Year Comparison",
    y = "No.Rainy Days",
  ) +
  theme_bw() +
  theme(
     axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.title.x = element_blank()
  )
```

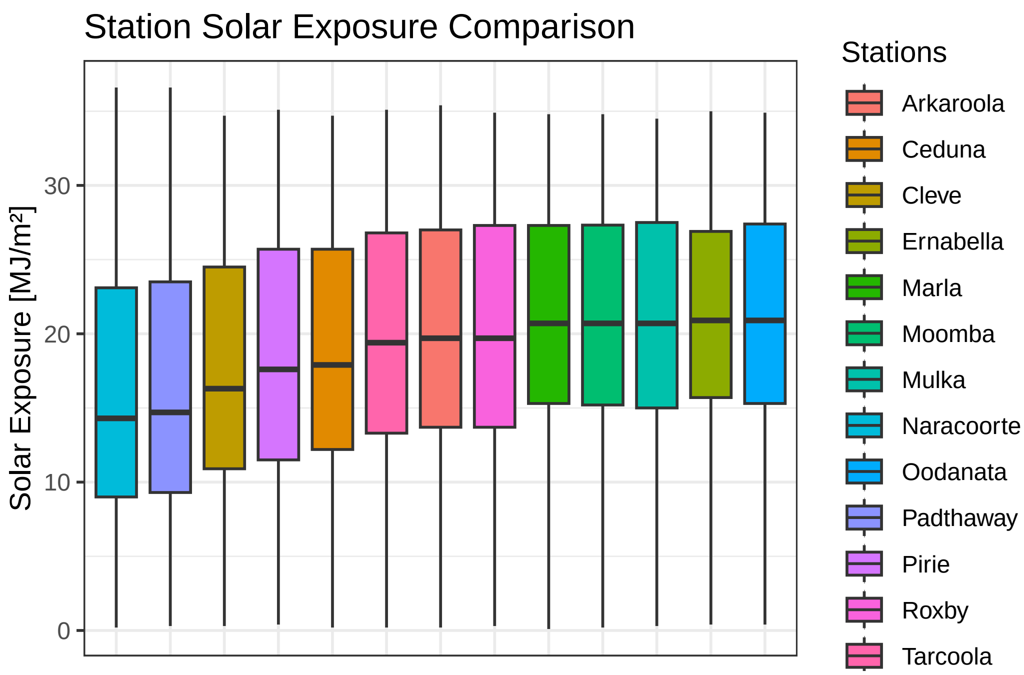


```
ggplot(Rain_Plotting, aes(x = reorder(Stations, `Daily Rainfall amount`, FUN = median, na.rm
  geom_boxplot() +
  labs(
    title = "Yearly Rainfall amount comparison",
    y = "Rainfall [ml]",
  ) +
  theme_bw() +
  theme(
     axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.title.x = element_blank()
  )
```

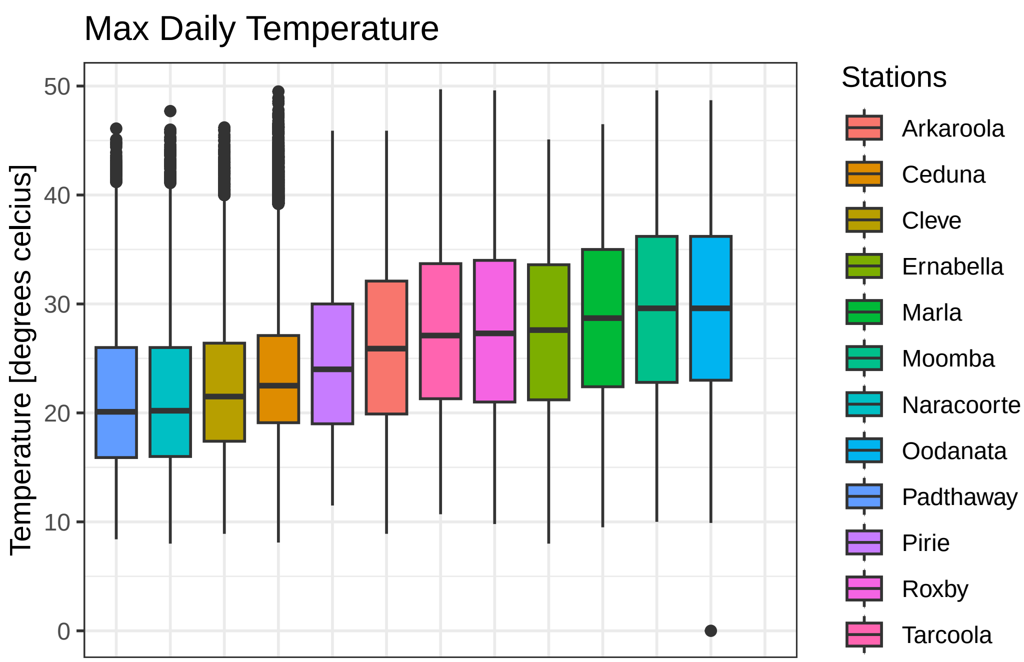



```
ggplot(Solar_Plotting, aes(x = reorder(Stations, `Solar Exposure Data`, FUN = median, na.rm =
  geom_boxplot() +
  labs(
    title = "Station Solar Exposure Comparison",
    y = "Solar Exposure [MJ/m²]",
  ) +
  theme_bw() +
  theme(
     axis.text.x = element_blank(),
```

```
    axis.ticks.x = element_blank(),
    axis.title.x = element_blank()
  )
```




```
ggplot(Temp_Plotting, aes(x = reorder(Stations, `Max Daily Temperature`, FUN = median, na.rm
  geom_boxplot() +
  labs(
    title = "Max Daily Temperature",
    y = "Temperature [degrees celcius]",
  ) +
  theme_bw()+
  theme(
     axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.title.x = element_blank()
  )
```

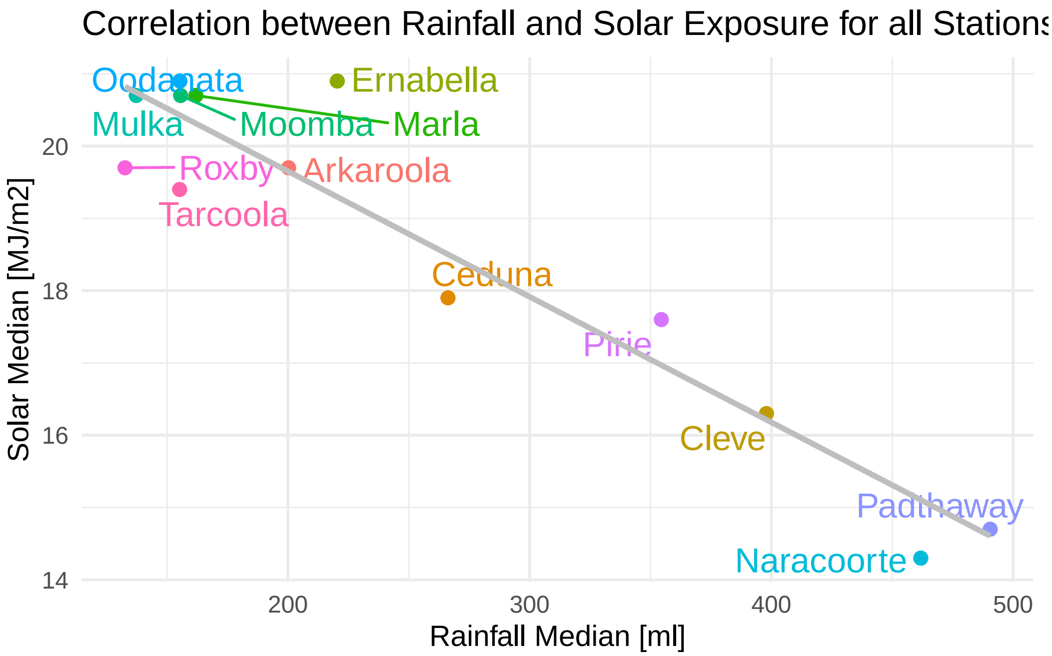


Scatter-plots

This code displays the scatterplots for all variable combinations

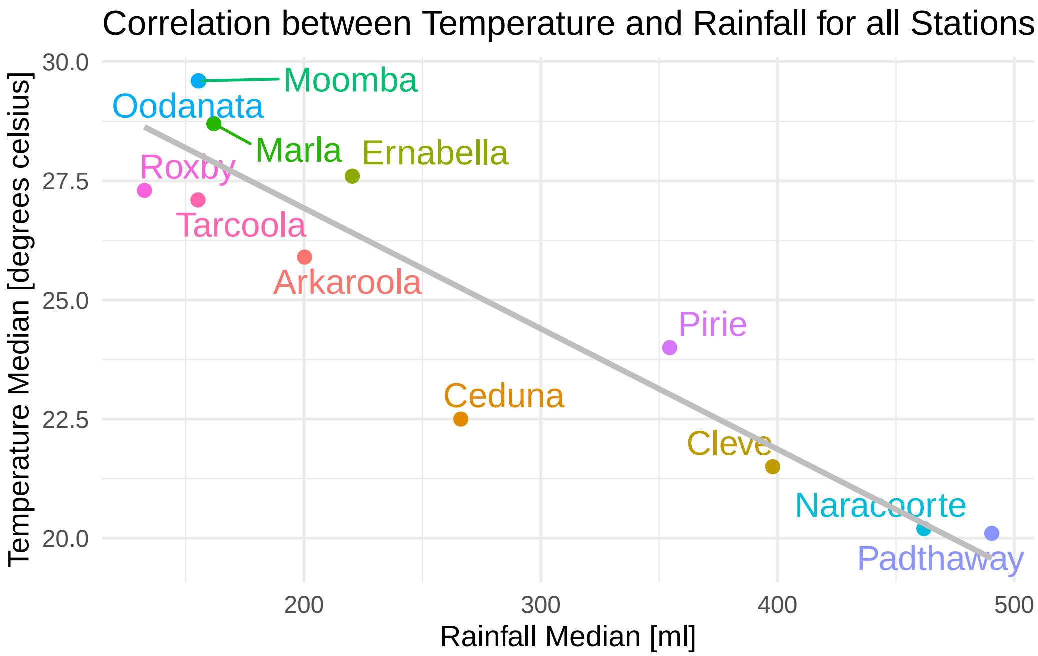
```
ggplot(median_values, aes(Rain, Solar, label = Location, colour = Location)) +
  geom_point(size = 2) +
  geom_text_repel(size = 4.5) +
  theme_minimal() +
  theme(legend.position = "none") +
  xlab("Rainfall Median [ml]") +
   geom_smooth(method = "lm", color = "grey", se = FALSE) +
  ylab("Solar Median [MJ/m2]") +
  ggtitle("Correlation between Rainfall and Solar Exposure for all Stations")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



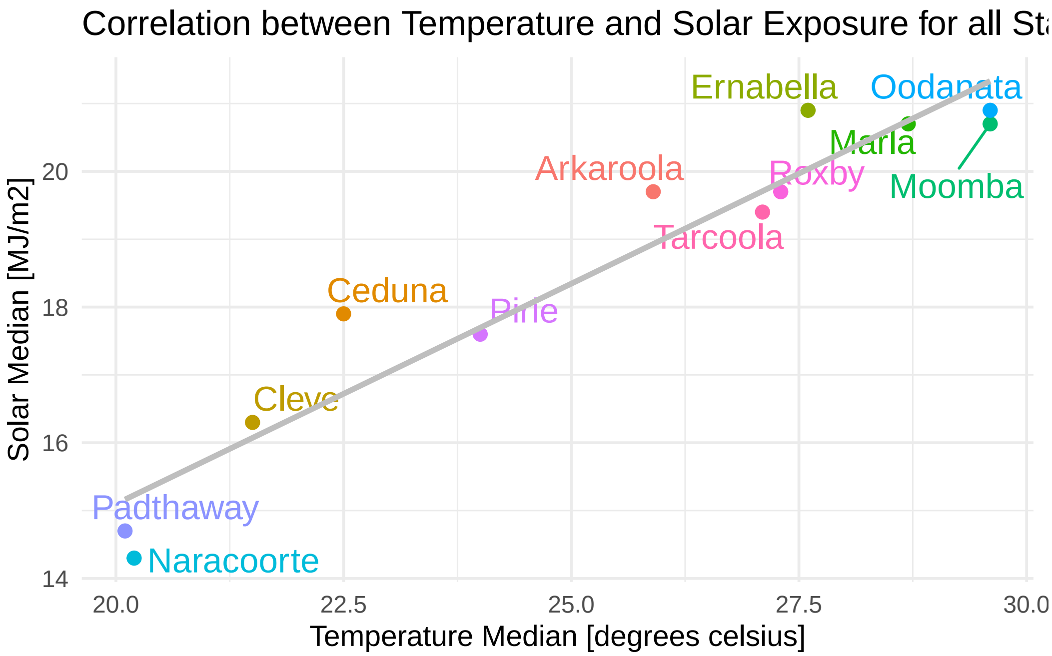

```
ggplot(median_values, aes(x = Rain, y = Temp, label = Location,colour=Location)) +
  geom_point(size = 2) +
  geom_text_repel(size = 4.5) +
  theme_minimal() +
  theme(legend.position = "none") +
  xlab("Rainfall Median [ml]") +
   geom_smooth(method = "lm", color = "grey", se = FALSE) +
  ylab("Temperature Median [degrees celsius]") +
  ggtitle("Correlation between Temperature and Rainfall for all Stations")
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
ggplot(median_values, aes(x = Temp, y = Solar, label = Location,colour=Location)) +
  geom_point(size = 2) +
  geom_text_repel(size = 4.5) +
  theme_minimal() +
  theme(legend.position = "none") +
  geom_smooth(method = "lm", color = "grey", se = FALSE) +
  xlab("Temperature Median [degrees celsius]") +
  ylab("Solar Median [MJ/m2]") +
  ggtitle("Correlation between Temperature and Solar Exposure for all Stations")
```

```
`geom_smooth()` using formula = 'y ~ x'
```


Correlation between Temperature and Solar Exposure for all Sta
Solar Median [MJ/m2]
Temperature Median [degrees celsius]
20
18
16
14
20.0
22.5
25.0
27.5
30.0
Ernabella
Oodanata
Marla
Arkaroola
Roxby
Moomba
Tarcoola
Ceduna
Pirie
Cleve
Padthaway
Naracoorte